ACIS 498: Information Systems Capstone

Instructor Contact Information

Instructor: Amul Chapla

Email: Amul.Chapla@northwestern.edu. Most emails/messages usually responded within 24-36 hrs. It could be 48-72 hours when I am traveling for business.

Phone: (312) 404-6640

Links: https://www.linkedin.com/in/amulchaplaLinks to an external site.

Office Hours: By Appointment Only

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Biography

Amul Chapla is a distinguished AI technology expert with over 20 years of experience. Currently the Chief AI Architect at Microsoft, he has a decade of specialized experience in cloud computing, AI, and data analytics, providing technical and strategic leadership to large companies to drive AI-powered business innovations. He frequently speaks at conferences about leveraging Generative AI to drive business value. Amul's contributions have earned him multiple prestigious awards from Microsoft.

Since 2018, he has been an Adjunct Professor at Northwestern University, developing and teaching AI and Data Science courses focused on practical, industry-relevant skills in AI and ML. His curriculum leverages open-source tools, cloud services, and hands-on programming to offer a comprehensive learning experience. Passionate about applying his professional expertise in academia, he ensures students gain valuable, applied experience in AI and Data Science. He holds an MS in Computer Science from the Illinois Institute of Technology and a Data Science professional certification from Microsoft.

Teaching Assistant Contact Information

Teaching Assistant: Cesar Martinez

Email: cesar-martinez1@northwestern.edu

Course Description

This course provides experience in the development and delivery of a large-scale software application that solves a real-world problem. This will be accomplished through a managed capstone software project that will cover all aspects of the software development life cycle including (but not limited to): discovery, requirements, design, implementation, testing, technical documentation, and deployment. Students will learn how to research a real-world problem, evaluate industry trends that address the problem, and consequently propose and implement their own solution to the problem. To accomplish this, students will learn how to apply the skills they acquired through the various tracks of the Computer Information System program to deliver the project, which in turn will set them up for success in their professional careers. The capstone project provides the opportunity to apply knowledge and skills in the tracks of Data Science, Artificial Intelligence, Database and Internet Technologies, Project Management, Programming, Information System Security, Health Informatics, and/or other disciplines in the Computer Information Systems program. Depending on the scope and complexity of the project, students will work individually or in groups to deliver the solution.

Course Objectives

By the end of this course, you will be able to:

- Identify a real world business problem, explain why a solution is needed, and describe what value is created when the problem is solved.
- Propose multiple approaches to solve the problem by analyzing technology practices and trends in the industry.
- Evaluate and explain which proposed solution approach creates the maximum value.
- Create a project that will implement the optimal solution, plan and manage its activities, and define its major outcomes and success criteria.
- Identify risk factors to the project and propose how to mitigate them.
- Apply experience and previous knowledge acquired in the program to deliver a software product that solves the identified business problem.
- Produce professional, clear, and articulate documentation of the project throughout all the stages of the development that follows industry standards.
- Present the project in an organized and structured manner to a wider audience. Identify and plan for appropriate controls and checkpoints to govern and maintain the BC/DR plan.

Prerequisites

The earliest students may take this course is in the quarter of their final class. Students must have completed CIS 413, CIS 414 or MSDS 430, and CIS 417 prior to taking this course; they may not take CIS 498 alongside one of these core courses.

University Syllabus Standards

Students are responsible for familiarizing themselves with this information:
https://www.registrar.northwestern.edu/registration-graduation/northwestern-university-syllabus-standards.html

Student Support Services

AccessibleNU

This course is designed to be welcoming to, accessible to, and usable by everyone, including students who are English-language learners, have a variety of learning styles, have disabilities, or are new to online learning. Be sure to let me know immediately if you encounter a required element or resource in the course that is not accessible to you. Also, let me know of changes I can make to the course so that it is more welcoming to, accessible to, or usable by students who take this course in the future.

Northwestern University and AccessibleNU are committed to providing a supportive and challenging environment for all undergraduate, graduate, professional school, and professional studies students with disabilities who attend the University. Additionally, the University and AccessibleNU work to provide students with disabilities and other conditions requiring accommodation a learning and community environment that affords them full participation, equal access, and reasonable accommodation. The majority of accommodations, services, and auxiliary aids provided to eligible students are coordinated by AccessibleNU, which is part of the Dean of Students Office.

SPS Student Services

The Student Services advising team assists students in a variety of matters during their time at SPS. Each program has an assigned adviser to help students with academic planning (course selection, degree planning, adding/dropping courses), policies, and administrative procedures, and to serve as a guide to resources at SPS and the greater Northwestern community. Student advisers use both proactive advising to keep students on track and intrusive advising to help resolve issues or concerns.

Student advisers can serve as a resource in student issues including, but not limited to, providing guidance to a student in danger of failing class or who may be looking to drop a class, making contact with a student who is not responsive to messages from faculty, and assisting a student experiencing family or medical issues that are impacting their academics. The advising team can be reached at spsacademicadvising@northwestern.edu, and more information on the academic advisers and available student services can be found on the SPS Student Services page.

Academic Support Services

Northwestern University Library

As one of the leading private research libraries in the United States, Northwestern University Library serves the educational and information needs of its students and faculty as well as scholars around the world. Visit the Library About page for more information or contact Distance Learning Librarian Tracy Coyne at 312-503-6617 or tracy-coyne@northwestern.edu.

Remote access to the library databases and many other university services requires Multi-Factor Authentication (MFA).

Program-Specific Library Guides

- Business Databases A-Z
- Data Management
- Data Science
- Information Systems

Additional Library Resources

- Connectivity: Campus Wireless and Off-Campus Access to Electronic Resources
- Getting Available Items: Delivery to Long-Distance Patrons
- Quick Access to Major Newspapers
- Reserve a Library Study Room
- Resources for Data Analysis
- Sign up for an in-person or online Research Consultation Appointment
- Social Science Data Resources

Required and Optional Readings and Resources

Required Readings

Adedeji, O. (2023). *Full-stack flask and react: Learn, code, and deploy powerful web applications with flask 2 and react 18*. Packt Publishing.

Additional required readings and media are posted on Canvas, including timely news articles, academic research, and videos that you will review in order to complete some assignments and participate in discussion forums.

Course Reserves

Some readings will be available through the Course Reserves in the course navigation menu. For assistance with Course Reserves, e-mail e-reserve@northwestern.edu. To ask a librarian for assistance, visit Northwestern's Ask A Librarian page.

Assignment Overview and Grading Breakdown

Module	Assessment	Description	Points
1	**Self-Evaluation of Technical & Management Skills Survey (UG)**	For this assignment, you will participate in a survey in which you will rate your technical and management skills to help the instructor form well-balanced groups. This self-assessment will earn you 2.5 points upon completion by the due date.	10/1%
1	**Proposal/Business Case Document:**	Each group or individual will submit a document that clearly explains the business problem/opportunity that the project will solve and how it aligns with broader business and/or technology strategy at a selected organization	20/2%
2	**Detailed Requirements Analysis Document**	Each group or individual will submit a document that provides a comprehensive evaluation of requirements that are relevant to the project scope. This document will be the **contract** between you and the instructor upon which grading will be based on at project delivery.	20/2%

<table>
<tr>
<td>9</td>
<td>Production Support Document

Test Scenarios

Complete Code from Programming Labs</td>
<td>Generate a production support document that contains the following:

1. Description of the application
2. Standard operating procedures
3. Troubleshooting steps to potential problems
4. How to perform a clean restart of your application if it goes down

Please use the provided template to create test scenarios to all your functional and non-functional requirements. It is expected that you provide more than 20 test scenarios for this assignment.

Submit all your code for the project (Python programs, HTML, Java Script, database DDLs, queries, test files, and any file containing code that is readable) in a zip file.</td>
<td>20/5%</td>
</tr>
<tr>
<td>9</td>
<td>Complete Project Documentation</td>
<td>Submit final project documentation that includes all the previous documents (except status reports) with any revisions due to instructor feedback or any modifications you introduced in the course of development. Make sure that you also submit your final test results of the Software product.

You are expected to provide clear and articulate writing that could be used in a professional environment.</td>
<td>50/5%</td>
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9	**Issue Diagnosis, research, resolution and sharing**	This project is focused on building practical skills for end-to-end full stack application development. Any practical application will have inherent issues during development, testing, deployment & maintenance stages. Each group will be required to follow below guidelines throughout the course to build the essential skills of issue identification, issue research and issue resolution approach. 1. Work as a group or individual to use internet search, ChatGPT, AI Copilots, Stack Overflow & other relevant resources to help with any technical issue. 2. If you can't resolve any issue as a group then post a detailed issue description on the course discussion board. Your post MUST include the steps you have already tried as a group to try to resolve the issue. Any issue post that does not include what steps you have tried as a group, what resource you have tried to use etc will result in 10 points deduction for the group. 3. If you encountered any issue and resolved it as an individual or group then post it on the discussion board highlighting issue and resolution details. Your group will get 5 bonus points for each issue with resolution post.	50/5%
3	**Capstone Project Milestone 1 solution**	Submit working code, with a video recording of the milestone 1 project solution **(Front-end component)**	100/15%

5	**Capstone Project Milestone 2 solution**	Submit working code, with a video recording of the milestone 2 project solution **(Back-end component – API with DB integration)**	100/15%
7	**Capstone Project Milestone 3 solution**	Submit working code, with a video recording of the milestone 3 project solution **(Front-end and Back-end components integrated).** Minor integration issues, errors are allowed. Must present a document with a list of known issues, feature gaps and plan to address them.	100/15%
8	**Capstone Project Milestone 4 End-to-end Dry-run**	Presentation and Complete Project solution demonstration)	100/10%
9	**Final Presentation (Final Grading)**	Your presentation is expected to be organized and well-structured and each group member(s) is expected to demonstrate knowledge of their part in the project as well as the project as a whole. The presentation should cover all required functionalities in a logical manner. Group members are expected to answer all questions effectively and speak fluently with confidence. Satisfaction of requirements is **worth 100** points of the total points.	250/25%
10	**Capstone Presentation**	You will apply feedback from the presentation delivered in the previous module.	50/5%
		Total	1000/100 %

Grading Scale

Grade	Points
A	94-100
A-	90-93
B+	87-89
B	84-86
B-	80-83
C+	77-79
C	74-76
C-	70-73
F	Below 70

The School of Professional Studies does not award D grades in graduate coursework.

Attendance & Late Work Policy

Attendance Policy

Regular, on-time attendance is mandatory. Students must be present and on time for each class session to be counted as "present." Arriving late will result in being marked "absent" for that day.

Class may begin with a quiz or activity assigned at the scheduled start time. If you are late, you will not be allowed to take the quiz or make it up later, resulting in a loss of points for that assessment.

Excessive absences or tardiness may negatively impact your final grade and your eligibility to remain in the course.

Late Assignment Policy

Assignments must be submitted on time. Late submissions will not be accepted unless prior written permission is granted by the instructor before the due date. Extensions will be approved only under exceptional circumstances, and documentation may be required. Plan ahead and manage your time effectively. Technical issues, travel, or conflicting deadlines in other courses are not valid reasons for extensions.

Use of Generative AI

Students are permitted—and encouraged—to use AI tools such as ChatGPT, GitHub Copilot, and similar assistants to support programming tasks, including writing code, debugging, and exploring solutions. However, use of these tools does not replace the responsibility to understand your code.

You are expected to fully comprehend any code you submit: how it works, why it works, and what it does. Assignments, quizzes, and exams may include questions that require you to explain or modify your code without AI assistance. Misuse of AI tools—such as copying code without understanding or submitting AI-generated content you cannot explain—will be treated as an academic integrity issue.

Online Communication and Interaction Expectations

Online Communication Etiquette

Beyond interacting with your instructor and peers in discussions, you will be expected to communicate by Canvas message, email, and sync session. Your instructor may also make themselves available by phone or text. In all contexts, keep your communication professional and respect the instructor's posted availability. To learn more about professional communication, please review the Connecting with Faculty guide.

Just as you expect a response when you send a message to your instructor, please respond promptly when your instructor contacts you. Your instructor will expect a response within two business days. This will require that you log into the course site regularly and set up your notifications to inform you when the instructor posts an announcement, provides feedback on work, or sends you a Canvas message. For guidance on setting your notifications, please review How do I manage my Canvas notification settings as a student? It is also recommended that you check your @u.northwestern e-mail account regularly, or forward your @u.northwestern e-mail to an account you check frequently.

SPS Learning Studios

Learning studios are available to students who would like additional support in commonly used tools and topics, including: statistics, and coding in R. An instructor is available to answer your questions as you work through self-paced content and exercises. Students can self-enroll for free by visiting the SPS Academic Services page.

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Read&Write

Read&Write is an optional text reading and writing program with numerous beneficial features. Originally developed to assist users with print disabilities, such as visual impairments, dyslexia, ADHD, etc., this program provides a wide array of tools to assist with reading, writing, and notetaking. One of the most useful tools is the text-to-speech function, which students may use to convert digital text into an audio format.

Read&Write is available for free to all Northwestern students, faculty, and staff. Visit the Northwestern IT site on Read&Write for more information about the software, as well as instructions on how to download it.

Course Technology

This course will involve a number of different types of interactions. These interactions will take place primarily through the Canvas system. Please take the time to navigate through the course and become familiar with the course syllabus, structure, and content and review the list of resources below.

Systems Requirements for Distance Learning

Students and faculty enrolled in SPS online classes should have access to a computer with the Minimum System Requirements.

Canvas

The Canvas Student Center includes information on communicating in Canvas, navigating a Canvas course, grades, additional help, and more. The Canvas at Northwestern website provides information on getting to know Canvas at Northwestern and getting Canvas support. The Canvas Student Guide provides tutorials on all the features of Canvas. For additional Canvas help and support, you can always click the Help icon in the lower left corner to begin a live chat with Canvas support or contact the Canvas Support Hotline.

The Canvas Accessibility Statement and Canvas Privacy Policy are also available.

Panopto

Videos in this course may be hosted in Panopto. If you have not used Panopto in the past, you may be prompted to login to Panopto for the first time and authorize Panopto to access your Canvas account. You can learn more about using Panopto and login to Panopto directly by visiting the Panopto guide on the Northwestern IT Resource Hub. Depending on the assignment requirements of this course, you may be asked to create videos using Panopto in addition to viewing content that your instructor has provided through Panopto.

The Panopto Privacy Policy and the Accessibility Features on Panopto are also available.

Turnitin

Assignments in this course are analyzed for similarity to web resources and past assignments via the Turnitin Plagiarism Framework as part of an automatic plagiarism review. When submitting an assignment that uses the Turnitin Plagiarism Framework, you will be prompted to acknowledge that you are submitting your own work and agree to the Turnitin End User License Agreement. The Canvas Plagiarism Framework Student guide page provides information about how the results of the Turnitin similarity report can be used by your instructor to enhance their ability to evaluate your submission. Turnitin provides additional information about accessibility and user privacy.

Technical Help and Support

The SPS Help Desk is available for Faculty, Students and Staff to support their daily IT needs. For additional technical support, contact the Northwestern IT Support Center

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Permissions

Instructional Materials

This course was developed in partnership with Distance Learning staff in the School of Professional Studies at Northwestern University. Every effort has been made to responsibly acquire instructional materials for this class, by adhering to copyright law, obtaining permission from copyright holders, selecting Open Educational Resources (OERs) and Creative Commons (CC) materials, and using citations to credit the work of others.

The same is expected of students in this course. Please review the Academic Integrity statement for more information.

Sharing Course Content

Content within this course--including assignment descriptions, exam questions, and other course components--may not be distributed outside of the course, either to other students or on the Internet more broadly.

Student Ownership of Content

Students retain ownership of all content developed while completing this course, as dictated by the university Copyright Policy ("copyright ownership resides with the Creator(s) of copyrightable works").

Per the Family Educational Rights and Privacy Act (FERPA), if your instructor wishes to share your work with future students, your permission must be obtained in writing.

Your instructor may limit access to the course after a cutoff date. When you complete the course, please ensure that you have saved all work. You may not be able to return to the course to download your submissions.

Course Schedule

Module 1: Introduction to Web Development and Front-End Basics (Project Discovery & Group Formation)

Learning Objectives

- Set up basic tools: VSCode, Git and GitHub repository.
- Explain the basics of web development, including the client-server model and HTTP/HTTPS protocols.

- Utilize HTML and CSS to create and style web pages.
- Experiment with anchor, image, list and other elements to construct web pages.
- Understand the core concepts of multi-page websites and computer file paths.
- Demonstration of an end-to-end website and a reference business problem.
- Plan and gather requirements for the capstone project.
- Identify real-world business problems for capstone project and explain why a web application development solution would solve the business problem.

Readings & Media

Required Readings

- Adedeji, O. (2023). *Full-stack flask and react: Learn, code, and deploy powerful web applications with flask 2 and react 18*. Packt Publishing.
    - Chapter 1

Assignments

- Programming Lab 1
- Module 1: Proposal/Business Case Document

Module 2: Introduction to JavaScript and DOM Manipulation (Project Proposal and Requirements)

Learning Objectives

- Summarize JavaScript concepts including Syntax, Variables, Functions, DOM Manipulation.
- Write basic JavaScript code for DOM manipulation.

- Explain the core concepts of React.js, including components, JSX, and props.
- Build React JavaScript applications that handle user events and state.
- Make use of React Router to build single-page applications (SPAs).

- Decide the final business problem that you will solve with a project plan to build a web application.

Readings & Media

Required Readings

- Adedeji, O. (2023). *Full-stack flask and react: Learn, code, and deploy powerful web applications with flask 2 and react 18*. Packt Publishing.
    - Chapters 2-3

Assignments

- Programming Lab 2
- Module 2: Detailed Requirements Analysis Document

Module 3: Building Dynamic React JavaScript Front-End Applications (Development activities)

Learning Objectives

- Set up and structure a React JavaScript application project.
- Understand and use APIs to fetch data in React applications.

- Explain what JSON is and how it is used in web development.
- Develop a hands-on project, such as a Todo List application, using React.
- Debug and test React applications effectively.
- Build the frontend part of the capstone project using React.

Readings & Media

Required Readings

- Adedeji, O. (2023). *Full-stack flask and react: Learn, code, and deploy powerful web applications with flask 2 and react 18*. Packt Publishing.
    - Chapters 4-7

Assignments

- Programming Lab 3
- Assignment Module 3: Capstone Project Management Assignment Milestone 1 - Front-End Component

Module 4: Back-end Basics and building RESTful APIs with Flask (Solution Architecture)

Learning Objectives

- Explain RESTful API concepts and use Flask-Restful to build endpoints.
- Outline the core concepts and architecture of Flask for API development.

- Build RESTful APIs with Flask & Python to perform CRUD operations
- Construct and manage routes for RESTful APIs in Flask.
- Explain API authentication and security concepts.
- Create solution architecture for the capstone project MVP.
- Build and test the front-end web component for the capstone project.

Readings & Media

Required Readings

- Adedeji, O. (2023). *Full-stack flask and react: Learn, code, and deploy powerful web applications with flask 2 and react 18*. Packt Publishing.
    - Chapter 9

Assignments

- Programming Lab 4

Module 5: Introduction to PostgreSQL Database for full-stack app (Full-stack Development)

Learning Objectives

- Explain the role of database in full-stack application development.
- Utilize PostgreSQL database to store, retrieve and manipulate data for web app

- Summarize the basics of databases and run SQL queries against PostgreSQL.
- Create database models in Flask applications.
- Test CRUD operations using PostgreSQL database.
- Solution full-stack application design utilizing front-end, back-end & database components.
- Develop the backend part of the capstone project using Flask.

Readings & Media

Required Readings

- Adedeji, O. (2023). [*Full-stack flask and react: Learn, code, and deploy powerful web applications with flask 2 and react 18*](). Packt Publishing.
    - Chapter 8,9, and 15

Assignments

- Programming Lab 5
- Module 5: Capstone Project Management Assignment 2

Module 6: Database Integration with back-end RESTful APIs (Continue Full-stack Development)

Learning Objectives

- Integrate Flask RESTful APIs with a relational database using SQLAlchemy.
- Understand database models and ORM (Object-Relational Mapping).
- Implement CRUD operations with a database backend through the back-end APIs.

- Explain options to secure API endpoints with authentication and authorization.
- Test REST APIs using API testing tools such as Postman.

Readings & Media

Required Readings

- Adedeji, O. (2023). *Full-stack flask and react: Learn, code, and deploy powerful web applications with flask 2 and react 18*. Packt Publishing.
    - Chapter 8,9,15

Assignments

- Programming Lab 6

Module 7: Full Stack Development - Connecting Front-end and Back-end components (Test Full-stack application)

Learning Objectives

- Integrate front-end web app with back-end via REST APIs

- Test front-end interactions with the database layer via REST APIs

- Install and set up CORS in Flask applications.
- Fetch data from Flask APIs in React applications.
- Implement Google OAuth for authentication (optional).
- Handle errors and optimize performance in full stack applications.

Readings & Media

Required Readings

- Adedeji, O. (2023). *Full-stack flask and react: Learn, code, and deploy powerful web applications with flask 2 and react 18*. Packt Publishing.
  - Chapter 10-11

Assignments

- Programming Lab 7
- Module 7: Capstone Project Management Assignment 3

Module 8: Capstone Project Checkpoint & Live Dry-run 1 (Capstone Solution Checkpoint 1)

Learning Objectives

- Test and debug end-to-end working for the capstone project.
- Revise and implement error handling in the front-end and back-end components as per test results.

- Perform live demonstration of the capstone project from student laptop. Note that cloud deployment is not required for Live Dry-run 1.
- Deploy the capstone project using Azure/AWS environment.

Readings & Media

Required Readings

- Adedeji, O. (2023). *Full-stack flask and react: Learn, code, and deploy powerful web applications with flask 2 and react 18*. Packt Publishing.
  - Chapter 13

Assignments

- Programming Lab 8
- Module 8: Capstone Project Management Assignment 4 End-to-End Dry-Run

Module 9: Capstone Project Presentation Draft & Live Dry-run 2 (Capstone Solution Checkpoint 2)

Learning Objectives

- Perform live demonstration of the capstone project with the solution deployed to a cloud environment.
- Dry-run of the presentation using the draft deck

Assignments

Module 9: Final Presentation

Module 10: Capstone Project Presentation & Application Demo (Capstone Presentation)

Learning Objectives

- All solution components/code must be finalized.
- Presentation deck must be finalized and polished.

Assignments

Module 10: Capstone Presentation